Using HTMX To Make Interactive Elements In Drupal

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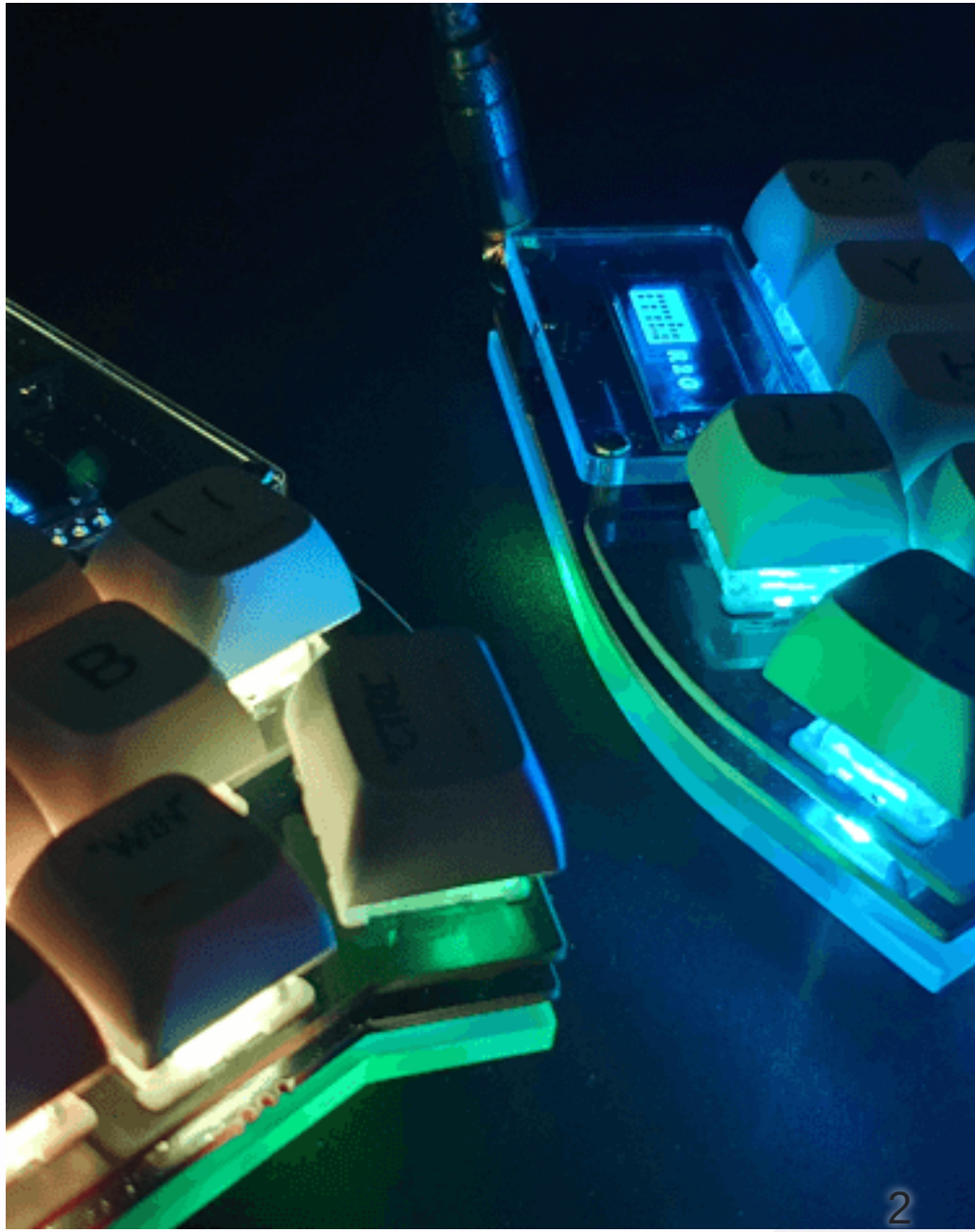

Source Code

- Talk is available at: https://github.com/hashbangcode/drupal-htmx-talk or via QR code
- All code seen can be found at:
  - https://bit.ly/46arnzj HTMX examples
  - https://bit.ly/3OHAIIR Drupal examples
- More resources at www.hashbangcode.com

Using HTMX To Make Interactive Elements In Drupal

HTMX In Drupal

- HTMX added to Drupal in 11.3 as a core component.
- This is the new standard for ajax requests.
- All existing ajax features will be re-written in HTMX.

HTMX

What is HTMX?

- JavaScript framework.
- Allows ajax calls and CSS Transitions *without writing JavaScript code*.
- Small at 16K (minified and gzipped).
- No other dependencies.
- Platform agnostic.
- Plugins extend functionality.

What is HTMX?

- Behaviour is added to HTML attibutes.
- Any element can issue a web request.
- All responses should be in HTML.

Installing HTMX

Download and include the single JavaScript file.

```
<script src="htmx.min.js"></script>
```

There are CDN options available.

We are now ready to use HTMX.

HTMX Attributes

- Prefixed with `hx-` or `data-hx-`.
- 11 attributes are commonly used.
- There are another 20-30 that add more functionality.

HTMX Attributes - Example

```
<button hx-get="index.php" hx-target="#div1">Submit</button>

<div id="div1"></div>
```

This example will:

- `hx-get` = Send a "get" request to the index.php path.
- `hx-target` = Write the response in the element with the ID "div1".

HTMX Requests

- All requests from HTMX will have the `hx-request` header.
- Certain attributes will add further headers to the request.

Responding To The Requset

Your response should be in plain HTML.

We could respond to the previous request with:

```
<p>Button clicked!</p>
```

Which would be injected into our page like this.

```
<div id="div1"><p>Button clicked!</p></div>
```

Responder Class In PHP

A simple class to respond to HTMX requests.

```php
class Htmx {
    public static function isHtmxRequest(): bool {
        return isset($_SERVER['HTTP_HX_REQUEST']) && $_SERVER['HTTP_HX_REQUEST'] === 'true';
    }

    public static function isGet(): bool {
        return $_SERVER['REQUEST_METHOD'] === 'GET';
    }
}
```

Responding To The Request

We detect the `hx-request` header and the correct HTTP method before responding.

```
if (Htmx::isHtmxRequest() && Htmx::isGet()) {
    echo '<p>Button clicked at ' . date('r') . '.</p>';
}
```

The div now looks like this:

```
<div id="div1"><p>Button clicked at Sun, 08 Feb 2026 16:00:23 +0000.</p></div>
```

HTMX Common Attributes

HTTP Verbs

Different HTTP verbs are available through attributes.

- `hx-get` does a "get" request.
- `hx-post` does a "post" requset.
- `hx-delete` does a "delete" requset
- `hx-patch` does a "patch" requset
- `hx-put` does a "put" requset

hx-target

The target element to be swapped.

```
<button hx-get="index.php" hx-target="#div1">Submit</button>

<div id="div1"></div>
```

hx-trigger

Defines the action that will trigger the request.

```
<div hx-post="index.php" hx-trigger="click">Click me</div>
```

- Can be left out for elements that have default triggers.
- Eg. `<button>` elements automatically have a `click` trigger.

hx-trigger Examples

```
<div hx-post="index.php" hx-trigger="click">Click me</div>
```

```
<div hx-post="index.php" hx-trigger="click once">Click me</div>
```

```
<div hx-post="index.php" hx-trigger="revealed"></div>
```

```
<div hx-post="index.php" hx-trigger="load delay:500ms"></div>
```

```
<input hx-get="index.php" hx-trigger="keyup delay:1s" />
```

hx-select

Select content to swap with the target element from the response.

HTMX:

```
<button hx-get="index.php" hx-select="#response" hx-swap="outerHTML">Click</button>
```

Response:

```
if (Htmx::isHtmxRequest() && Htmx::isGet()) {
    echo '<p id="response">Button clicked at ' . date('r') . '.</p>';
    echo '<p>Some extra content that won\'t get displayed.</p>';
}
```

hx-select-oob

Select an "Out Of Band" element to also target in the response.

HTMX:

```
<button hx-get="index.php" hx-select-oob="#div1">Send Request</button>
<div id="div1"></div>
```

Response:

```
if (Htmx::isHtmxRequest() && Htmx::isGet()) {
  echo 'Request Sent';
  echo '<div id="div1">Button clicked at ' . date('r') . '.</div>';
}
```

hx-select-oob - Continued

You can comma separate this property to do multiple things.

HTMX:

```
<button hx-get="index.php" hx-select-oob="#div1,#div2">Send Request</button>
<div id="div1"></div> <div id="div2"></div>
```

Response:

```
if (Htmx::isHtmxRequest() && Htmx::isGet()) {
  echo 'Request Sent';
  echo '<div id="div1">Button clicked at:</div>';
  echo '<div id="div2">' . date('r') . '.</div>';
}
```

hx-swap

- Controls how an element will be swapped into the page.
- Defaults to `innerHTML` - replace the HTML contents.
  - `outerHTML` - Replace the element.
  - `textContent` - Replace contents without parsing as HTML.

hx-swap - Continued

- `beforebegin`, `afterbegin`, `beforeend`, `afterend` - Place the response before or after the element and its children.
- `delete` - Delete the target element from the page.
- `none` - Do nothing with the response (but still process out of band items).

hx-swap-oob

Swap an "Out Of Band" element. Similar to `hx-select-oob` but the response tells HTMX what to swap.

HTMX:

```
<button hx-get="index.php" hx-swap="none">Send Request</button>
<div id="div1"></div>
```

Response:

```
if (Htmx::isHtmxRequest() && Htmx::isGet()) {
    echo '<div id="div1" hx-swap-oob="true">Button clicked at ' . date('r') . '.</div>';
}
```

CSS Transitions

- HTMX adds classes to elements during different events.
- `htmx-added` is added to new content before it is added to the page.
- We can use these classes to animate what HTMX is doing with CSS transitions.

CSS Transitions

The following will fade in a bit of content inside the `div1` element as it is loaded into the page.

```
#div1 p {
    opacity: 1;
    transition: opacity 1s ease-in;
}
#div1 p.htmx-added {
    opacity: 0;
}
```

Configuring HTMX

- It is possible to configure HTMX via a number of settings.
- Can be set through meta tags (preferred)

```
<meta name="htmx-config" content='{"defaultSwapStyle":"outerHTML"}'>
```

Or JavaScript.

```
<script>
htmx.config.defaultSwapStyle = 'outerHTML';
</script>
```

DEMO!

HTMX In Drupal

HTMX In Drupal

- HTMX is included in Drupal as a library.
- A few classes exist to assist with adding attributes and responding to requests.

HTMX in Drupal

The standard usage of HTMX in Drupal is to use the `data-hx-` prefix for attributes.

Eg:

```
<button data-hx-get="index.php" data-hx-target="#div1">Submit</button>

<div id="div1"></div>
```

Drupal Libraries

HTMX can be included into the page through a Drupal library.

- `core/htmx` - The core HTMX library.
- `core/drupal.htmx` - Additional scripts for Drupal.
- You rarely need to inject these libraries directly.

Drupal Libraries

The library `core/drupal.htmx` includes three files:

- `htmx-assets.js` - Adds assets the current page requires.
- `htmx-behaviors.js` - Connect `Drupal.behaviors` to HTMX inserted content.
- `htmx-utils.js` - Helper functions for the other two files.

Drupal Integration

- The `\Drupal\Core\Htmx` class is used as a wrapper around the HTMX libraries and attribute injection for render arrays.
  It can also inject headers for responding to HTMX.

- `\Drupal\Core\Htmx\HtmxRequestInfoTrait` is used to detect HTMX requests and associated headers.

Htmx Class

- Has a number of methods that set up attributes.

```php
$output['element'] = [
  '#type' => 'html_tag',
  '#tag' => 'p',
  '#value' => 'Content'
];

$htmx = new Htmx();
$htmx->get()
  ->swap('afterend')
  ->trigger('revealed')
  ->applyTo($output['element']);
```

```html
<p
  data-hx-get="/route"
  data-hx-swap="afterend"
  data-hx-trigger="revealed">
Content
</p>
```

Htmx Class

- The Htmx class can also be used with a different syntax.

```
(new Htmx())
  ->get()
  ->swap('afterend')
  ->trigger('revealed')
  ->applyTo($output['element']);
```

HTMX Drupal Response

- Drupal has the ability to respond to a HTMX response with HTML.
- This uses the class `Drupal\Core\Render\MainContent\HtmxRenderer`.
- Renders a little HTML document and sends this upstream.

HTMX Drupal Response

HtmxRenderer is invoked by:

- `_wrapper_format=drupal_htmx` as a query on the incoming request.
- The `_htmx_route=true` on the route reponding to the request.

HTMX With Controllers

HTMX With Controllers

- Think about your approach.
  - One route: with logic to separate out the HTMX response.
  - Two routes: one marked with `_htmx_route: TRUE` .

HTMX With Controllers

The `__htmx_route: TRUE` is used to create the HTMX response action.

```
mymodule_controller_action_htmx:
  path: '/htmx-response'
  defaults:
    _controller: '\Drupal\mymodule\Controller\MyModuleController::htmx'
  requirements:
    _permission: 'access content'
  options:
    _htmx_route: TRUE
```

HTMX With Controllers

The `HtmxRequestInfoTrait` trait needs access to the request stack service.

```php
namespace Drupal\mymodule\Controller;

use Drupal\Core\Controller\ControllerBase;
use Drupal\Core\Htmx\HtmxRequestInfoTrait;

class MyController extends ControllerBase {

  use HtmxRequestInfoTrait;

  public function __construct(protected RequestStack $requestStack) {}

  public function action() {}
}
```

HTMX With Controllers

Then, in your action:

```
public function action() {
  if ($this->isHtmxRequest()) {
    // Respond to HTMX request.
  }

  // Respond to normal controller action.
  // Generate markup to set up the HTMX request.
}
```

HTMX With Forms

HTMX With Forms

- The trait `HtmxRequestInfoTrait` is part of the `FormBase` class.
- The `request_stack` service is also part of the form.

HTMX With Forms

- The HTMX response from forms is the entire form, so you *need* to add a `hx-select` (or similar) to pick out the relevnat part of the response.
- Also, forms need to be consistent. You can't just throw elements into the form markup as the elements need to exist in the form build.

HTMX With Forms

- Forms in the same way as constructors, you just need to decorate the elements in question.

```
(new Htmx())
  ->post()
  ->target('*:has(>input[name="email"])')
  ->select('*:has(>input[name="email"])')
  ->trigger('keyup delay:1s')
  ->applyTo($form['email']);
```

- This decorates the form element with the HTMX elements.

DEMO!

Drupal Needs You!

- HTMX is in Drupal 11.3 as a fully featured system.
- All of the ajax subsystem now needs to be converted to HTMX.
- There's a lot of work to be done.
- https://www.drupal.org/community-initiatives/replace-ajax-api-with-htmx

HTMX Module

- https://www.drupal.org/project/htmx
- Some extra tools and examples of HTMX in a Drupal setting.
- Has a Views plugin to display Views as HTMX.
- Has a HTMX debugger that logs all HTMX events in the browser console.

Resources

- #htmx on Drupal Slack.
- htmx.org - https://htmx.org
- HTMX Labs - https://htmxlabs.com

Questions?

- Slides: https://bit.ly/4tFKUBK

Thanks!

- Slides: https://bit.ly/4tFKUBK